

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity. (BL2, BL3)*

Activity Tool : MCQ Test (Low)
Assessment Tool Weightage: 30%

QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	Which of the following is true about 1NF? (a) Allows multi-valued attributes (b) Eliminates partial dependencies (c) All attributes must be atomic (d) Eliminates transitive dependencies	BL2 – Understand	5
2	A relation is in 2NF if it is in 1NF and: (a) Has no transitive dependencies (b) Has no partial dependencies on primary key (c) Every attribute is a prime attribute (d) Has no multi-valued dependencies	BL2 – Understand	5
3	Which normal form deals with multi-valued dependencies? (a) 2NF (b) 3NF (c) BCNF (d) 4NF	BL2 – Understand	5
4	BCNF is a stricter version of: (a) 1NF (b) 2NF (c) 3NF (d) 4NF	BL2 – Understand	5
5	Functional dependency $X \rightarrow Y$ means: (a) Y determines X (b) X determines Y (c) X and Y are unrelated (d) Both determine each other	BL2 – Understand	5
6	Which of the following anomaly is NOT associated with unnormalized relations? (a) Insertion anomaly (b) Deletion anomaly (c) Update anomaly (d) Retrieval anomaly	BL2 – Understand	5
7	A superkey is: (a) A minimal set of attributes that uniquely identifies tuples (b) Any set of attributes that uniquely identifies tuples (c) An attribute that depends on a non-prime attribute (d) A foreign key reference	BL2 – Understand	5

8	If R(A,B,C) has FD: $A \rightarrow B$ and $B \rightarrow C$, then C is: (a) Partially dependent on A (b) Fully dependent on A (c) Transitively dependent on A (d) Independently determined	BL2 – Understand	5
9	5NF is also known as: (a) Domain-Key Normal Form (b) Boyce-Codd Normal Form (c) Project-Join Normal Form (d) Element Normal Form	BL2 – Understand	5
10	Lossless-join decomposition ensures: (a) No data is lost during decomposition (b) All FDs are preserved (c) The relation is in BCNF (d) No null values exist	BL2 – Understand	5

Code of Conduct:

*I Sree B (192524023) _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

DATABASE MANAGEMENT SYSTEMS

ASSESSMENT TOOL 3 – ANSWERS

Course Code: CSA05 | Course Outcome: CO3

1. Which of the following is true about 1NF?

- (a) Allows multi-valued attributes
- (b) Eliminates partial dependencies
- (c) All attributes must be atomic
- (d) Eliminates transitive dependencies

Answer: (c) All attributes must be atomic.

Explanation: First Normal Form (1NF) requires every attribute to contain only atomic, indivisible values. It does not eliminate partial or transitive dependencies.

2. A relation is in 2NF if it is in 1NF and:

- (a) Has no transitive dependencies
- (b) Has no partial dependencies on primary key
- (c) Every attribute is a prime attribute
- (d) Has no multi-valued dependencies

Answer: (b) Has no partial dependencies on primary key.

Explanation: A relation is in 2NF when it is already in 1NF and every non-prime attribute is fully functionally dependent on the whole primary key, so partial dependency is removed.

3. Which normal form deals with multi-valued dependencies?

- (a) 2NF
- (b) 3NF
- (c) BCNF
- (d) 4NF

Answer: (d) 4NF.

Explanation: Fourth Normal Form (4NF) addresses multi-valued dependencies and helps eliminate redundancy caused by independent multi-valued facts.

4. BCNF is a stricter version of:

- (a) 1NF
- (b) 2NF
- (c) 3NF
- (d) 4NF

Answer: (c) 3NF.

Explanation: Boyce-Codd Normal Form (BCNF) is stricter than Third Normal Form (3NF). In BCNF, every determinant of a non-trivial functional dependency must be a superkey.

5. Functional dependency $X \rightarrow Y$ means:

- (a) Y determines X
- (b) X determines Y
- (c) X and Y are unrelated
- (d) Both determine each other

Answer: (b) X determines Y.

Explanation: $X \rightarrow Y$ means that the value of X uniquely determines the value of Y. Therefore, whenever two tuples have the same X value, they must have the same Y value.

6. Which of the following anomaly is NOT associated with unnormalized relations?

- (a) Insertion anomaly
- (b) Deletion anomaly
- (c) Update anomaly
- (d) Retrieval anomaly

Answer: (d) Retrieval anomaly.

Explanation: Insertion, deletion, and update anomalies are common problems caused by redundancy in poorly designed or unnormalized relations. Retrieval anomaly is not one of the standard modification anomalies.

7. A superkey is:

- (a) A minimal set of attributes that uniquely identifies tuples
- (b) Any set of attributes that uniquely identifies tuples
- (c) An attribute that depends on a non-prime attribute
- (d) A foreign key reference

Answer: (b) Any set of attributes that uniquely identifies tuples.

Explanation: A superkey is any set of one or more attributes that uniquely identifies each tuple in a relation. A candidate key is a minimal superkey.

8. If $R(A,B,C)$ has FD: $A \rightarrow B$ and $B \rightarrow C$, then C is:

- (a) Partially dependent on A
- (b) Fully dependent on A
- (c) Transitively dependent on A
- (d) Independently determined

Answer: (c) Transitively dependent on A.

Explanation: Since $A \rightarrow B$ and $B \rightarrow C$, A determines C through the intermediate attribute B. Therefore, C is transitively dependent on A.

9. 5NF is also known as:

- (a) Domain-Key Normal Form
- (b) Boyce-Codd Normal Form
- (c) Project-Join Normal Form
- (d) Element Normal Form

Answer: (c) Project-Join Normal Form.

Explanation: Fifth Normal Form (5NF) is also called Project-Join Normal Form (PJ/NF).

It deals with join dependencies and ensures that a relation cannot be further decomposed without losing information.

10. Lossless-join decomposition ensures:

- (a) No data is lost during decomposition
- (b) All FDs are preserved
- (c) The relation is in BCNF
- (d) No null values exist

Answer: (a) No data is lost during decomposition.

Explanation: A lossless-join decomposition allows the original relation to be reconstructed exactly by joining the decomposed relations. It does not necessarily guarantee dependency preservation or BCNF.